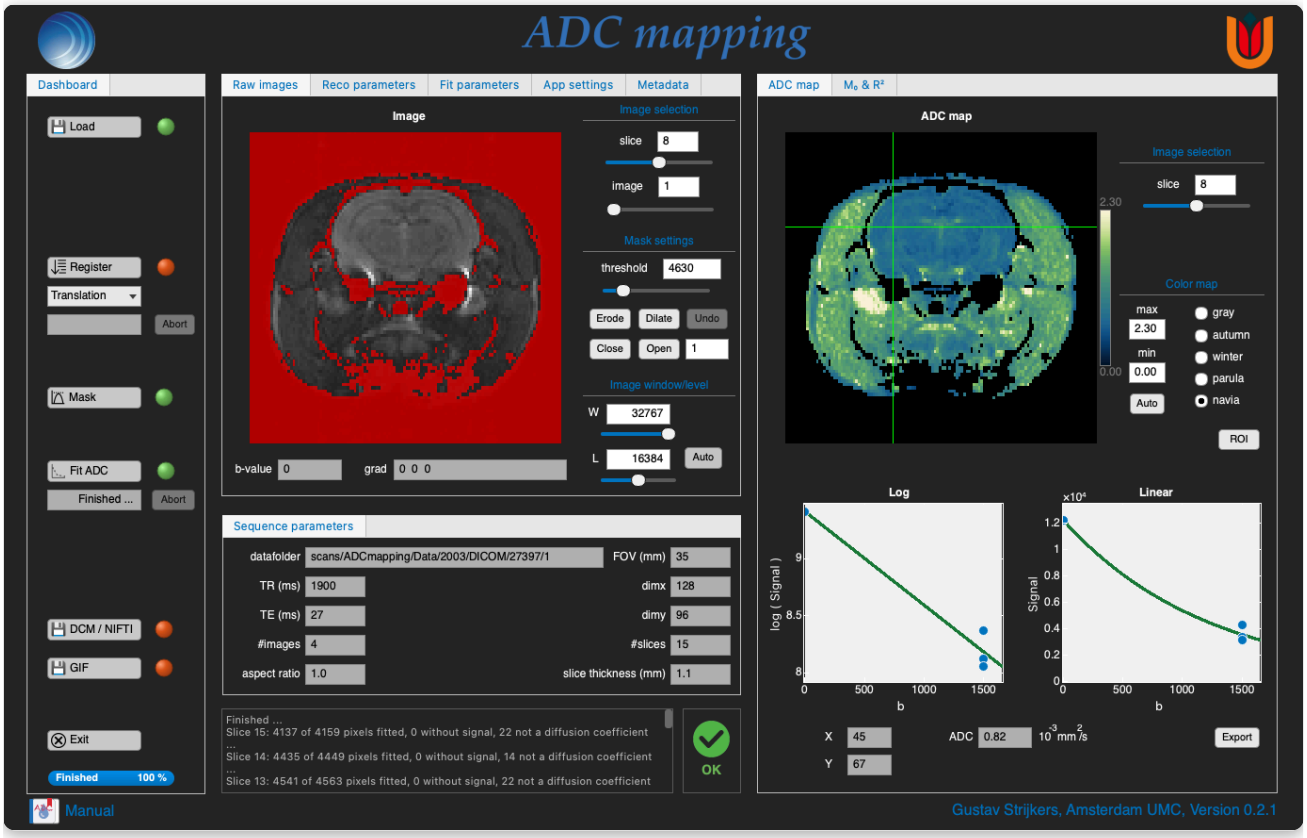



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## 1. Overview

ADCmapp is an application with a graphical user interface to reconstruct apparent diffusion coefficient (ADC) maps from diffusion-weighted acquisitions with an MR Solutions preclinical MRI system. It reads a diffusion-weighted EPI scan from its MRD files, which it reconstructs, or a series of DICOM files. The images are registered, masked and fitted to ADC, $M_0$ and $R^2$ maps, which can be exported as DICOM, NIfTI and GIF, with the values of a region of interest as a CSV file.

Note: the program does not calculate a diffusion tensor, but only estimates the mean diffusivity / ADC.

ADCmapp is a stand-alone application: everything it uses, elastix and dcm2niix included, is inside it, and nothing else has to be installed.

## 2. Contact

ADCmapp and this manual are written by:

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## 3. Disclaimer

This is a beta version. The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers or Bram Coolen.

## 4. Installation

### macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

ADCmapp comes as a disk image, `ADCmapp-0.2.5-macOS-arm64.dmg`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **ADCmapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying ADCmapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/ADCmapp.app
   ```

The first start can take a little longer: macOS checks the app once.

### Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

ADCmapp comes as an installer, `ADCmapp-0.2.5-setup.exe`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in `%LOCALAPPDATA%\Programs\ADCmapp` for the current user, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut. At the end it can start the app.

To remove the app, use **Uninstall ADCmapp** in the Start menu, or **Settings > Apps**.

### Elastix registration

Image registration uses elastix, which comes with ADCmapp. When elastix cannot be run, ADCmapp registers the images with a registration of its own instead (a B-spline registration is then done as affine), and says so.

ADCmapp does not use BART.

## 5. Running the software

Open **ADCmapp** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The ADCmapp logo appears while the app loads, and the window follows within seconds. At its start the app looks for elastix, and the message box reports what it found; "Ready" says it is done. Only **Load** and **Exit** are on until a dataset is loaded.

**Newer versions.** At its start the app asks GitHub (https://github.com/Moby1971/MRI-apps) whether there is a newer release of it. The message box then shows the current and the newest version and, when there is a newer one, where to download it. Without a network connection it says that the version could not be checked, and the app works as usual. `checkForUpdates` in the settings file switches the check off; `versionCheckTimeout` sets how many seconds it waits for GitHub (5).

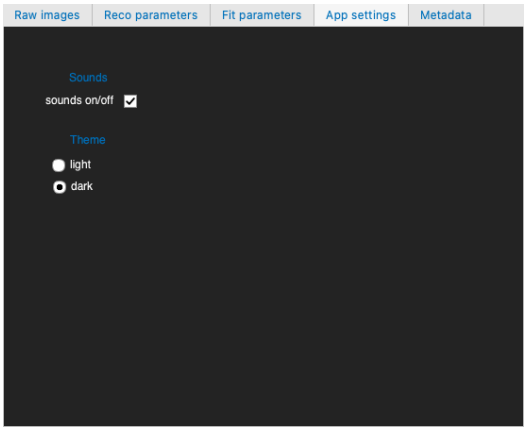
**This manual.** The **Manual** button at the bottom of the window, with the app's icon on a book, opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

### Settings and log files

ADCmapp keeps its files in a folder of its own, which it makes at its first start:

| | |
|---|---|
| macOS | `~/Library/Application Support/ADCmapp/0.2.5p` |
| Windows | `%APPDATA%\ADCmapp\0.2.5p` |




**Settings.** The settings are kept in `adcmappSettings.json`. The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds the version check, the folders where loading, the gradient table and the exports start (`importPath`, `gradientTablePath`, `defaultExportPath`), the theme, the sounds, the corrections of the MRD reconstruction, MP-PCA de-noising (`denoising.pca`, on, and `denoising.dicom`, off: also for DICOM series), the registration method, the $R^2$ cut-off and whether it is applied, the color maps of the three maps, and the number of log files kept. A value that cannot be used is replaced by its standard value, and the message box says so; a file that cannot be read at all is set aside as `adcmappSettings-unreadable-<time>.json` and replaced by a new one. The theme and the sounds can also be switched in the App settings tab, for the session only.

**Log files.** Every session writes a log file, `adcmapp-log-<date and time>.txt`, to the same folder. The log keeps every message and the parameters of each load, registration and fit. Logs older than the newest 50 are removed at start.

## 6. Basic operation




The ADCmapp program operates from a single window with 5 panels.

**Panel 1: Dashboard.** The task buttons and parameters that control the process. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed. At the bottom of the Dashboard one progress bar shows the progress of every task ("Loading 40 %", "Finished").

**Panel 2: Raw images / Reco parameters / Fit parameters / App settings / Metadata.** The last tab, **Metadata**, holds the record of the scan: the animal, the session and the contrast agent (step 8).

**Panel 3: Sequence parameters.** The relevant acquisition parameters, and the dataset loaded (**datafolder**).

**Panel 4: ADC map / $M_0$ & $R^2$.** During and after fitting, the ADC maps. The $M_0$ and $R^2$ goodness-of-fit maps are displayed in the second tab.

**Panel 5: Messages.** The program messages, newest on top, and the status: OK, BUSY, WARNING or ERROR.

### Step 1: Loading data

Press **Load** and choose a file:

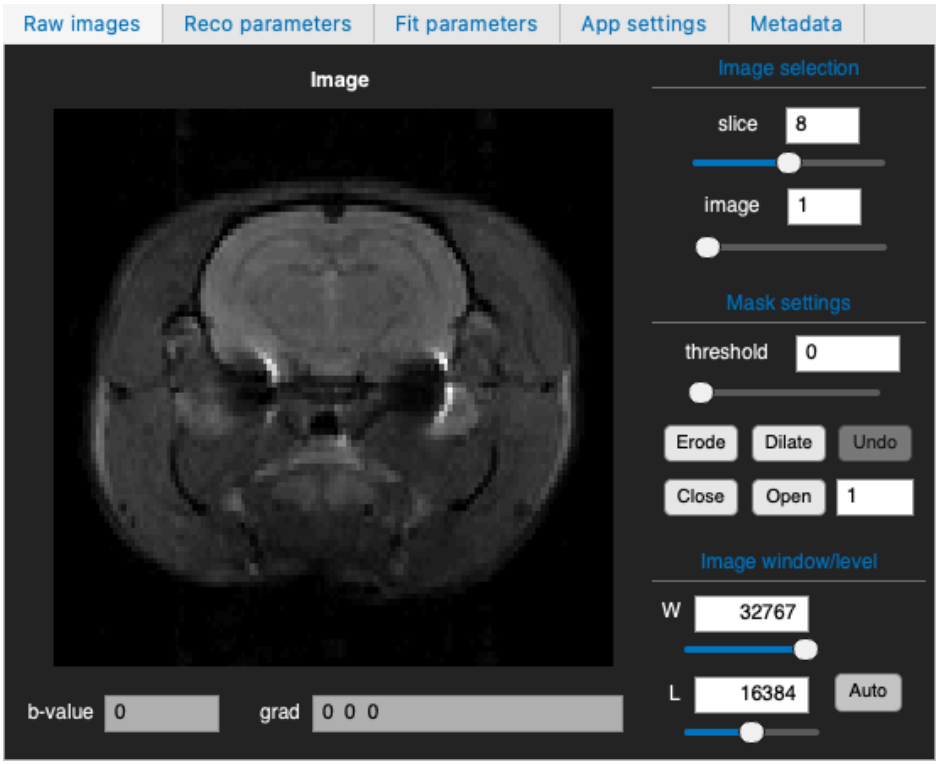
- an **MRD file** of a diffusion-weighted EPI scan, which is then reconstructed (step 2), or
- **any DICOM file of a series**: the series in its folder is loaded.

The middle slice and the first image are shown. A dataset that cannot be used is refused, with the reason in the message box, and the dataset loaded before stays. Reconstructions written by Retrospective or P2ROUD are not measurements and are refused: choose the scan's own MRD file.

The relevant acquisition parameters are shown in panel 3.



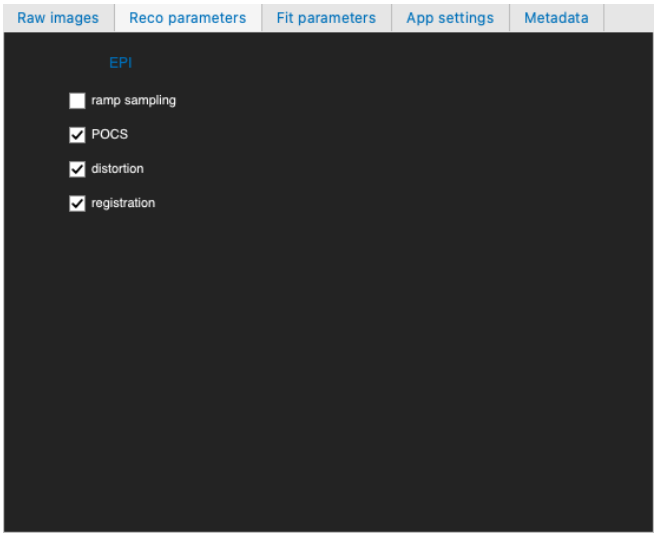

Images with different b-values and diffusion directions can be inspected with the sliders of the Raw images tab (**slice** and **image**); the **b-value** and gradient direction (**grad**) of the image are shown below it. The window and level are set with the **W** and **L** fields and sliders, or with **Auto**.




### Step 2: MRD scans

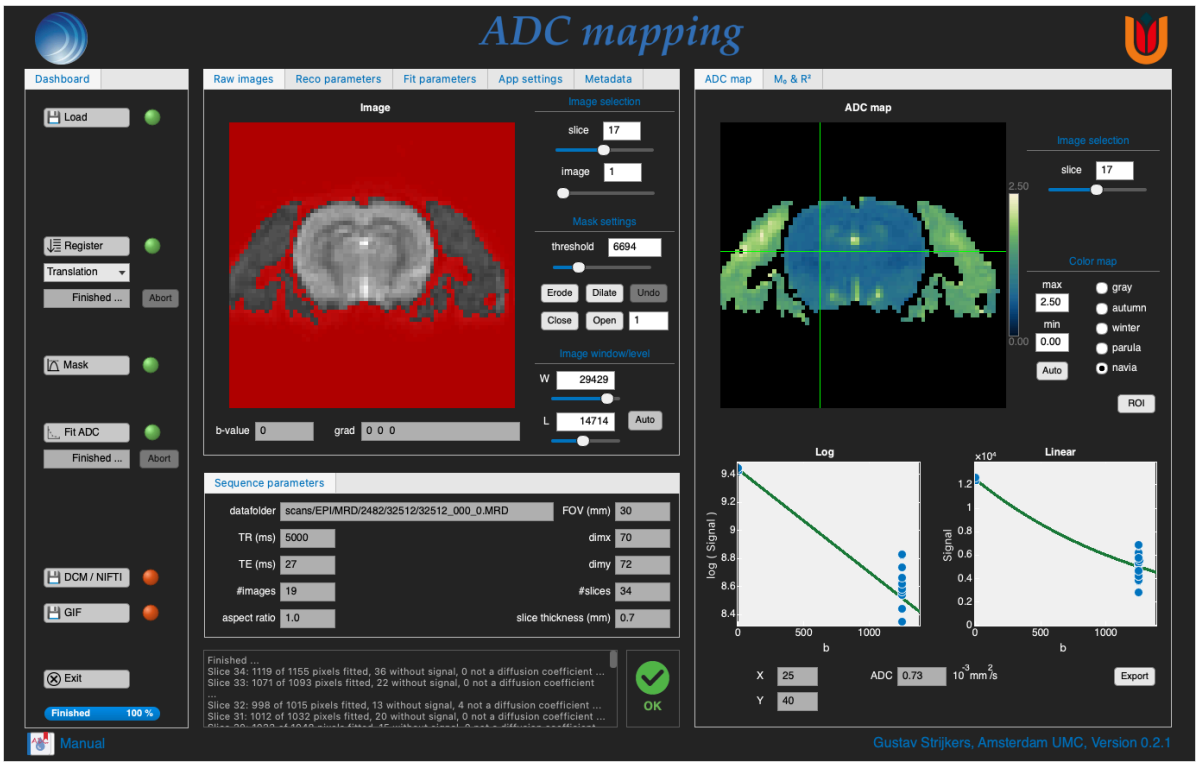
An MRD scan must be a 2D diffusion EPI with at least two b-values. Its MRD files are reconstructed at load, with the corrections of the Reco parameters tab:

- **ramp sampling** (off by default) — also uses the samples taken on the readout gradient ramps;
- **POCS** — completes a partial Fourier EPI, which sharpens the image and costs signal-to-noise ratio;
- **distortion** — the EPI geometric distortion is corrected, where the study holds a scan of the same volume with the phase encoding reversed;
- **registration** — the volumes of the series are aligned.



They start as the settings file says and take effect at the next load; MP-PCA de-noising is set in the settings file. The images are masked when the study has a FLASH mask scan. The reconstruction shows its progress on the progress bar; **Abort** (beside the registration) stops it, and the dataset loaded before stays.

Loading writes nothing beside your data: the reconstruction is kept in a private temporary folder, which the app removes again when the scan is exported, another scan is loaded, or the app closes. The **DCM / NIFTI** export writes the reconstruction (step 9). The window below shows a DTI scan of a brain, loaded from its MRD files, registered and fitted.




### Step 3: Image registration

NOTE: this step usually can be omitted.

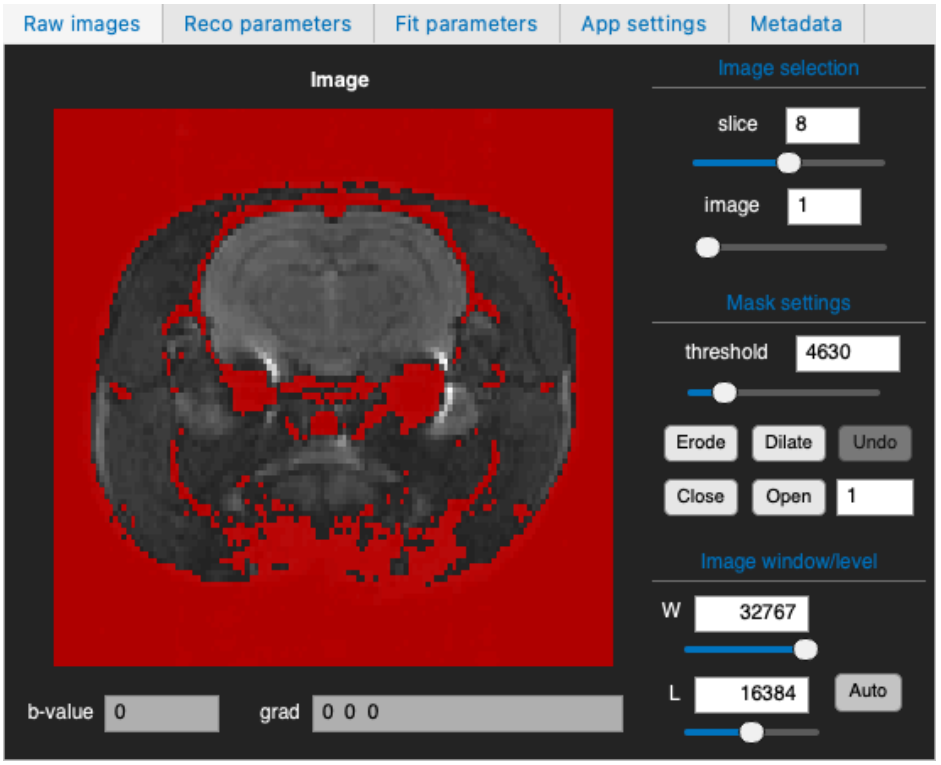
In case the images with different b-values and directions are not perfectly registered, press **Register** to register every image to the first image with the lowest b-value of its slice, with elastix, as a **Translation**, **Rigid**, **Affine** or **B-Spline** transform (the list below the button). **Abort** stops the registration; the images are then partly registered. Registering clears the mask and the maps. A series with many images takes a while: the DTI scan above, 19 images in 34 slices, took several minutes.

### Step 4: Masking the background

To prevent unnecessary fitting of the background pixels, the images need to be masked first; without a mask every pixel is fitted, the background included, with a warning.

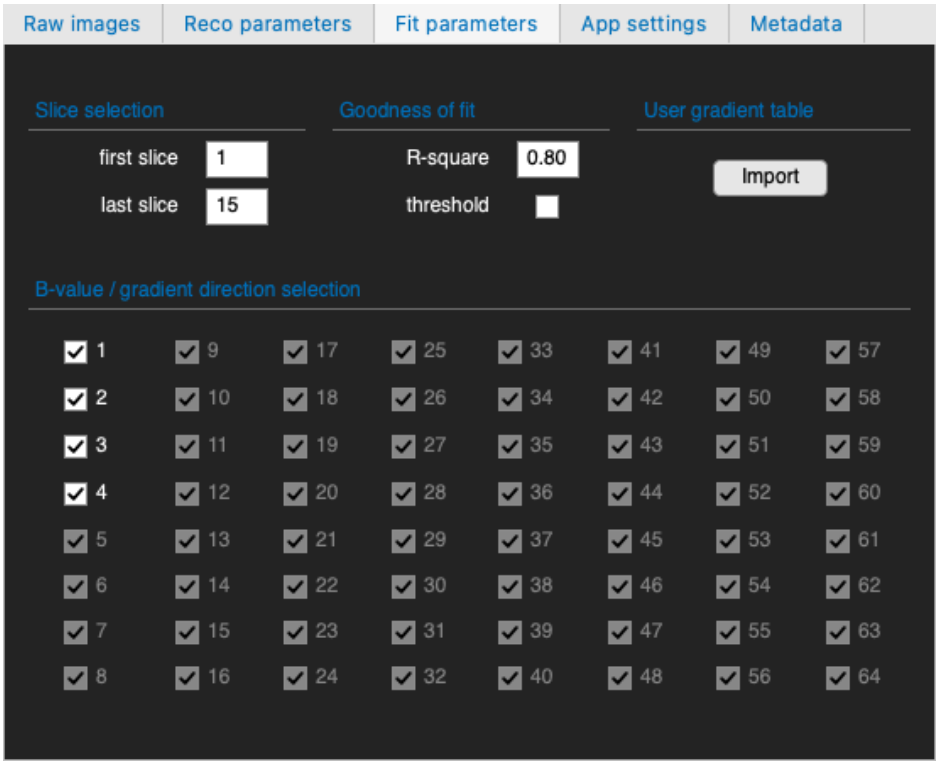
Press **Mask** to mask them with an automatic (Otsu) threshold per slice, on the first image with the lowest b-value. The red pixels indicate the regions that will be skipped during ADC fitting.

In case the automatic threshold is not satisfying, it can be adjusted per slice with the **threshold** field or slider. The mask can be refined by **Erode**, **Dilate**, **Close** and **Open** operations, of the size in pixels in the field beside them; **Undo** takes the last one back.




### Step 5: Fitting ADC maps

The Fit parameters tab holds the fit settings.



The first and last slice that will be fitted can be selected.

When the **threshold** box is ticked, the ADC and $M_0$ of a pixel whose $R^2$ is below **R-square** (0.80 by default) are cleared; the $R^2$ map keeps them, so the rejected pixels can be seen.

**$R^2$** measures how far the decay over the b-values departs from a mono-exponential; the differences between the diffusion directions of one b-value do not lower it.

$R^2$ needs at least 3 different b-values. With two — b = 0 and one shell, the usual trace or DTI acquisition — the line runs through the mean signal of both, so $R^2$ is 1 in every pixel and the threshold removes nothing; the message box says so when the fit starts.

The bottom part of this tab lists all the images (b-value and diffusion direction combinations). The images that are ticked are included in the fit.

**Import** reads a gradient table: a text file with a line per image, the b-value (s/mm²) and the three components of the direction, separated by spaces, tabs, commas or semicolons; % and # start comments. A dataset has to be loaded first; maps fitted before need to be fitted again to use the table.

Press **Fit ADC** to start the ADC fitting; **Abort** stops it. The maps fill in slice by slice. The selected b-values must take two values or more, at least 100 s/mm² apart. The ADC, in $10^{-3}$ mm²/s, is fitted to:

> $Signal = M_0 \exp(-b \cdot ADC)$

For each slice the message box says how many pixels were fitted.

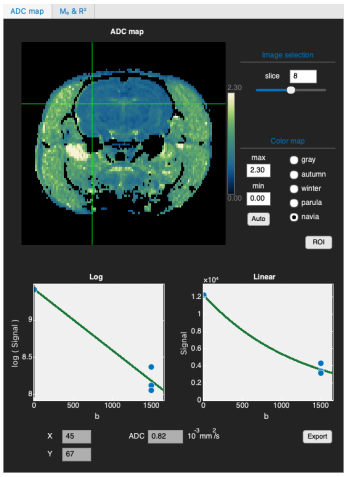
**The acquisition.** The fitted ADC is the mean diffusivity (the trace of the diffusion tensor over three) when each b-value is acquired with a balanced set of directions: three orthogonal directions, or a set spread evenly over the sphere, as a DTI scheme is. With other schemes the ADC leans toward the directions measured, and so does $R^2$; a scan meant for ADC maps should be planned that way.

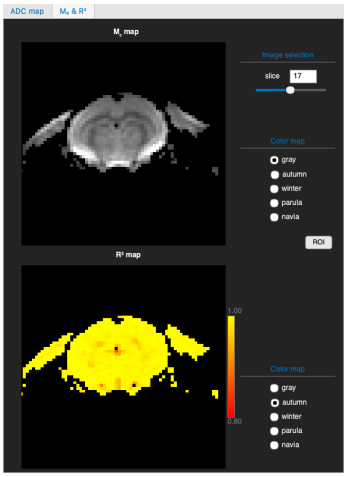
### Step 6: Fit results

The ADC maps are shown in the ADC map tab, on the scale set with **max** and **min**, or with **Auto** (0 to twice the mean ADC). An $M_0$ map and an $R^2$ map are shown in the second tab, each with its own color map (gray, autumn, winter, parula, navia).

The color bars stand beside the ADC and $R^2$ maps, with their end values above and below them: the scale's max and min for the ADC, 1 and the $R^2$ cut-off for $R^2$.

**ADC map** **$M_0$ & $R^2$, multi-shell DTI**



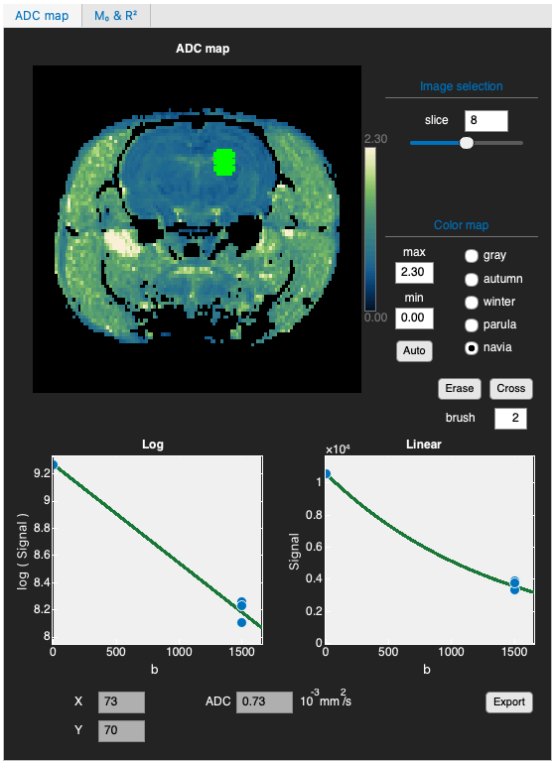




The $M_0$ & $R^2$ example is a multi-shell DTI scan (b = 0, 1000 and 2000 s/mm²): with three b-values $R^2$ shows where the decay is not mono-exponential.

### Step 7: ADC values of pixels and ROIs

When you click on the ADC map (or the $M_0$ map) a cross-hair is shown, with the log(signal) and the signal of that pixel as a function of b (symbols), the fitted curve (line) and its ADC. A red symbol indicates that this image was left out of the fit, according to the list in the Fit parameters tab (see step 5).

You can toggle between the cross-hair and region-of-interest (ROI) values with the **ROI** / **Cross** button. A single ROI can be 'painted' on the map, in one or more slices, by mouse clicks on the image, with the brush size in pixels set below the button. The plots then show the mean signal of the ROI and the curve of its mean ADC and $M_0$.

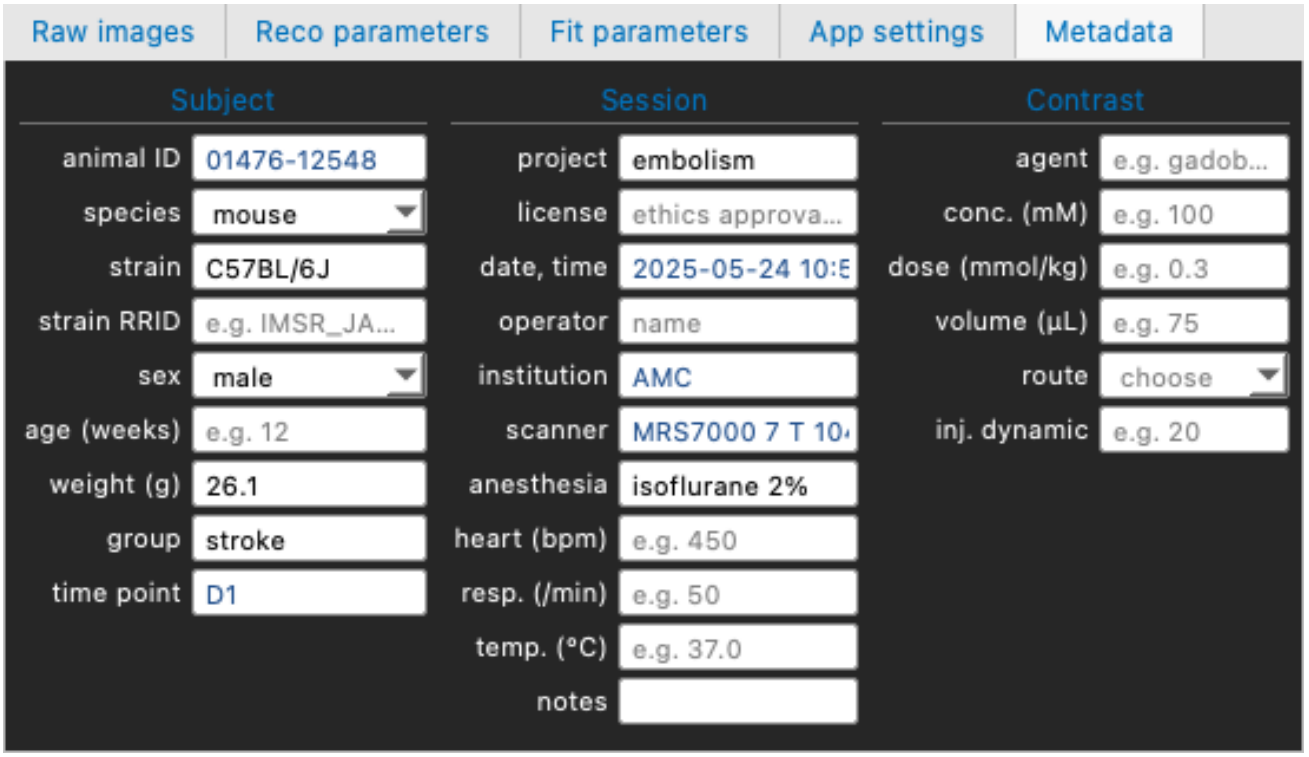


The ROI can be removed with **Erase**. **Export** writes the b-values and the signal under the cross-hair or of the ROI to a CSV file, `roiData_<tag>.csv`, in a folder you choose.

### Step 8: Metadata

The **Metadata** tab, beside the Raw images, Reco parameters, Fit parameters and App settings tabs, holds the record of the scan, which every export writes with the maps. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;
- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.




At a load the app fills in what the scan's files say. For a DICOM series that is its header: the animal ID and the time point from the patient name typed at the console (`20240212_926752_D0`: animal 926752, day 0), the project, the date and time, the institution and the scanner. For an MRD scan it is the `.rpr` file beside the MRD file, for the animal ID, the time point, the date and time, the institution and the scanner, and the scanner's DICOM images of the scan, where they are found, for the project. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty. A value the app filled in is shown in blue; every field can be corrected, and then counts as yours.

What was filled in for an earlier scan of the same exam is filled in again, from the records the apps wrote in the exam's `Metadata` folder (ADCmapp and the other apps alike) and from the tab as it was for the scan loaded before: the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature, the notes and the injection's dynamic belong to one scan and are never taken from another.

Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

### Step 9: Exporting the maps

There are two ways to export the ADC, $M_0$ and $R^2$ maps.

1. **DCM / NIFTI** — exports the ADC, $M_0$ and $R^2$ maps in DICOM and NIfTI format for further processing in third-party software, with a copy of the diffusion-weighted images. For an MRD scan it first writes the reconstruction: the DICOM series with `recoparameters_<tag>.txt`, its NIfTI files with the b-values and directions (`.bval`, `.bvec`), scripts for MRtrix3 and DSI Studio where the diffusion scheme allows it (written, not run), and a BIDS copy. The DICOM maps, and the reconstruction of an MRD scan, carry the species, strain, sex, age and weight of the animal and the contrast agent, where the Metadata tab has them.
2. **GIF** — exports the three maps as animated GIF files, one frame per slice, with two pictures of the app's window.

The exports go beside the scanner's files, in …`/Data/<study>/DICOM`, `Nifti`, `BIDS` and `GIF`, when the dataset was loaded from the scanner's folders; otherwise the app asks for a folder. Every export gets a new numbered folder, so that no export overwrites another.

Every export also writes, in …`/Data/<study>/Metadata/<tag>ADC/<n>`:

- `metadata.json` — the Metadata tab's record;
- `adcmapp-report_<tag>.json` and `.txt` — the report of the export: the app and its libraries, the scan and a checksum of its files, its acquisition with the b-values and directions, every step of the session with its parameters (the reconstruction of an MRD scan, the gradient table, the registration, the mask and your changes to it, the fit with all its settings), the statistics of each map per slice, the b-values the fit used, and every file the export wrote with its checksum;
- `ro-crate-metadata.json` — the same as an RO-Crate (https://www.researchobject.org/ro-crate/), a standard description of the folder for data repositories.

The paths in them are relative to the study folder, so they hold no names of your own folders.

### Step 10: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while a step runs: the step is stopped, the window disappears, and the program closes as soon as the step has ended. The temporary folder of an MRD scan is removed.

## 7. Credits

- **elastix** — S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, elastix: a toolbox for intensity based medical image registration, IEEE Transactions on Medical Imaging 29(1), 196–205 (2010). https://elastix.dev
- **PCA denoising** — J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, E. Fieremans, Denoising of diffusion MRI using random matrix theory, NeuroImage 142, 394–406 (2016).
- **Color maps** — F. Crameri, Scientific colour maps (navia), Zenodo, doi:10.5281/zenodo.1243862.
- **NIfTI conversion** — dcm2niix: C. Rorden et al., https://github.com/rordenlab/dcm2niix.